1.1 (a)

There was a substitution gene mutation where A was substituted by C in the code 2 and A in the last code was substituted by C; ✓Al This led to formation of changed codons on mRNA during transcription which attracted changed anticodons on tRNA during translation leading to change in amino acid sequence of the protein formed; ✓Ae sequence of amino acids changed leading to formation of nonfunctional fibrous proteins in cardiac muscle cells causing poor contraction and heart problems, irregular heart beat and weak heart muscles and caused the muscle to become stiff; ✓Ap

Average number of cristae on mitochondria was lower in diseased muscle tissue than in healthy heart tissue; ✓Al because of higher levels of Arsenic; ✓Ae Arsenic damages DNA sequence in region that controls formation of MICOs the Complex proteins for folding the inner membrane of the mitochondria. Such protein wrong protein were formed due to changed amino acid sequences causing poor folding of the inner membrane; ✓Ap

Mitochondrial oxygen consumption was lower in Diseased muscle tissue; ✓Al because of lower average number of cristae; ✓Ae This reduced surface area for electron carriers in the ETS this reduce the rate of movement of electrons and usage of oxygen in the mitochondria since oxygen is the last electron acceptor in the chain causing low energy available for muscle contraction; ✓Ap

(b) Regular Cardiac exercise will be effective; ✓El because Exercise stimulate Cardiac muscle to Synthesize more Mitochondria increasing surface area for by cristae and ATP production for the available actin fibres increasing strength of Contraction. However since the heart muscle is already stiff and Cannot Contract properly Intense exercise can cause micro-tears in the scar tissue causing low cardiac output which can cause shortness of breath and fainting; ✓Ee Therefore Continue with this method but only do light exercise for a short period; ✓Ep

Gene therapy will be effective; ✓El because it will Correct the mutation that by introducing a function gene leading to formation of correct actin and myosin proteins leading to proper Contraction However gene to be introduced may not easily reach in all cells; ✓Ee Therefore continue with this strategy but repeat it Several times to reach in many cells; ✓Ep

Being injected with oxygen-rich glucose Intravenous Supplements will be effective;

✓El because glucose will provide a respiratory substrate for glycolysis and oxygen will provide more electron acceptor making the few available cristae to make ATP More ATP improving muscle contraction However over supply of glucose Can cause hyperglycaemia and since arsenic levels are still high and DNA is damaged wrong proteins will keep being formed and symptoms will reappear once infusion Stops; ✓Ee Therefore continue with this method but use it only when symptoms appear to reduce symptoms and Combine it with reducing arsenic levels; ✓Ep

1.2 (a) Cell membrane permeability to glucose was lower in a diabetic patient than healthy individual; ✓Al due to low levels of Insulin in blood in diabetic patients;

✓Ae Since Insulin hormone is required for the to attach to receptors on cells to signal opening of channel protein that allow entry of glucose into the cell, low levels of insulin stimulate too few channels in the membrane to open leading to low permeability of cells to glucose; ✓Ap

OR

There was lower glucose uptake in diabetic patients than healthy individuals; ✓Al due to low level of Insulin in blood; ✓Ae This leads to low binding of glucose insulin to receptors on cells and many gated channel protein fail to open to allow in glucose into the cells; ✓Ap

Fibroblast mitotic Index % in diabetic patients was lower than normal; ✓Al due to low glucose uptake in cells; ✓Ae low glucose in cells provides few substrates reducing rate of ATP formation needed by fibroblasts to undergo active processes like synthesis of new material like spindle fibres and DNA replication resulting into slower mitotic rate this cause low fibres in blood vessels leading to hardening; ✓Ap

Time taken for the wound to heal in muscle cells was longer in diabetic patients than in Healthy individual; ✓Al because of low intracellular ATP in foot cells; ✓Ae ATP is needed to synthesis collagen fibres to close the wound and for cell division to regenerate the lost tissue so tissues will be formed slowly leading to higher time to heal; ✓Ap

Intracellular ATP concentration in diabetic patients was lower than normal; ✓Al because of low glucose uptake/ because of High blood glucose level; ✓Ae This makes blood viscous and narrow make blood vessel narrow and stiff reducing

supply of blood to far extreme tissues like foot leading to low oxygen and glucose supply needed to make glucose ATP in foot cells; ✓Ap

(b) Use of recombinant insulin from bacteria to treat type 1 diabetic patients will be effective; ✓El because Recombinant Insulin has same structure to the human insulin and will increase insulin in blood stimulating glucose uptake and reducing the sugar levels in blood without causing allergic reactions. However the inserted

Insulin stops working after some time leading to rise in blood sugar; ✓Ee therefore Continue with this strategy but Inject the insulin every 30 minutes before a meal; ✓Ep

Islet Cell transplant; ✓El This Introduces Beta cells that make insulin that triggers cells to convert glucose to glycogen lowering blood sugar level and solving the root cause However transplanted islets can easily be attacked /rejected by the patient's immune system; ✓Ee Therefore continue with this strategy but take drugs to suppress the immune system; ✓Ep

Carry out simple regular exercises; ✓El This will increase glucose usage by the cells in respiration to produce ATP lowering blood sugar naturally However if the exercise is intense muscles can use a lot of glucose leading to hypoglycaemia; ✓Ee Therefore engage in simple routine exercise; ✓Ep

1.3 (a) Mutation frequency per cell cycle control genes was higher in factory exposed workers than rural community; ✓Al Because of exposed to poisonous gases from the industries; ✓Ae these gases contain chemical that can affect arrangement of nucleotides in nucleic acids causing mutations resulting to cancer n lungs liver damage and abnormal growth; ✓Ap

Amount of ATP in cells of exposed factory workers was lower than in rural community; ✓Al because Hydrogen cyanide exposure from factories in the industrial area; ✓Ae Cyanide in body of the exposed factory workers reduced ATP production since it inhibits cytochrome oxidase inhibiting transfer of electron to electron Oxygen. Low levels of ATP resulted into fatigue in the workers; ✓Ap

Viscosity of Mucus in Factory workers was higher than in rural community; ✓Al because of High exposure to heavy metal D in Industrial smoke; ✓Ae this attached to another part of an enzyme making it irreversibly change shape reducing water movement into mucus in respiratory tract. This made mucus to be abnormally

thick blocking air ways and causing persistent stimulating coughing reflex and causing breathing difficulties; ✓Ap

Rate of abnormal division was higher in exposed workers than in rural community;

✓Al due to higher mutation frequency in cell cycle control genes; ✓Ae those genes control production of proteins that regulate cell cycle. Mutations in those genes resulted in change in mRNA formed during transcription and changed amino acid sequence during translation leading to formation of changed proteins. As a result cell cycles were not controlled resulting into abnormal division in lungs liver and skin causing Cancer in these areas; ✓Ap

(b) Install precipitators and filters on factory Chimneys; ✓Al This will trap smoke containing heavy metal D and reduce Inhalation by the factory workers However HCN is gas and can still pass through filter and precipitators; ✓Ee Continue with this strategy but pass the remaining gas in alkaline solution of NaOH to trap Hydrogen Cyanide that escapes the filter; ✓Ep

Factory workers should use personal protective equipments like Respirator with a mask filter; ✓El This will trap particles of heavy metals and protect workers against inhaling the gas however they can cause breathing difficulty increasing fatigue; ✓Ee Therefore continue with this strategy but make worker to work forget regular breaks between work; ✓Ep

plant a trees around the factory; ✓El Trees absorb Hydrogen cyanine released and enzyme inhibitor reducing release to the waste sources However trees can transfer these chemical to animals through food chain; ✓Ee continue with this strategy but plant non edible trees ; ✓Ep

1.4 (a) The boy had substitution gene mutation where G in code 3 was replaced by A; ✓Al This led to change of codon on mRNA during transcription attracting a changed tRNA anticodon during translation carrying a different amino acid leading to formation of a changed protein instead of cytochrome oxidase; ✓Ae faulty enzyme formed could not transfer catalyze transfer of electrons to oxygen reducing ATP formation during ETS leading to fatigue and poor growth; ✓Ap

Average number of ATP production per mitochondria was lower in the boy than normal; ✓Al due to faulty complex IV; ✓Ae this resulted into accumulation of electrons stopping ATP production during Electron transport system since this

enzyme is needed to catalyze transfer of electrons to oxygen leading long time for the wounds to heal and constant falling during walking; ✓Ap

Rate of mitosis in damaged tissues was lower than normal; ✓Al due to low ATP production; ✓Ae ATP is needed in mitosis to synthesize new material like spindle fibres and replication of organelles and DNA, low ATP delayed these processes leading to low Rate of division. Cells divided slowly to replace the damaged cells in the wounds causing longer time for the wound to heal; ✓Ap

Transitional epithelium of the boy had pores instead of intact membrane; ✓Al because of low ATP production; ✓Ae ATP is required during protein synthesis to synthesize proteins required to form the basement membrane and cell to cell junctions due to low ATP less of these proteins were made making transitional epithelial cells not to stick tightly together leading to soft bladder that leaks to urine; ✓Ap

(b) Gene therapy will be effective; ✓El because it will introduce a functional gene into cells of the boy leading to synthesis of a correct complex IV restoring ATP production to normal However the new gene may not reach all cells; ✓Ee Therefore continue with this strategy but repeat it several times for the gene to reach most cells; ✓Ep

Take coenzyme supplements; ✓El coenzyme will support aerobic respiration leading to production of ATP improving energy availability reducing fatigue and poor growth However it does not correct the mutation and symptoms keep on reappearing; ; ✓Ee Therefore continue with this strategy but only use it to treat symptoms; ✓Ep

Taking sodium bicarbonate solution ; ✓El this will neutralize acidity of blood as a result of lactic acid accumulation and prevent the blood from becoming too acidic; However excessive bicarbonate can react with stomach producing large quantities of carbon dioxide gas that causes bloating;✓Ee therefore continue with this strategy but only tae small quantities periodically; ✓Ep

1.5 (a) John's CFTR gene contains a base substitution (CTT to TTT) a mutation; ✓Al which alters the amino acid sequence and tertiary structure of the CFTR protein, producing a non-functional protein that affects chloride ion movement. The wrong CFTR gene transcribes a wrong mRNA because of the substituted (wrong) base TTT. This mRNA moves out of the nucleus to the rough endoplasmic

reticulum. The mRNA attaches to the ribosomes then tRNA transfer amino acids to the mRNA. The faulty base code then attaches to a corresponding tRNA carrying a different amino acid from the normal one. A ribosome in translation produced a protein with a different sequence of amino acids. Once translated, the formed protein chain enters the endoplasmic reticulum to be folded into its complex shape. Because of the wrong sequence of amino acids, the protein folds incorrectly; ✓Ae the wrong protein is destroyed hence no chloride channels for the cell. Chloride ions cannot move out of the airway epithelial cells, so no osmotic gradient forms and less water enters the mucus, producing thick, dehydrated mucus that accumulates in the airways Mucus in the narrowed airways causes turbulent airflow, producing wheezing, while the accumulated mucus stimulates the cough reflex, causing a persistent cough; ✓Ap

Blocked airways reduce oxygen uptake, leaving less oxygen available for aerobic respiration and lowering ATP production (as indicated by low mitochondrial activity). Less ATP is produced than the metabolic needs of John's body hence feeling fatigue; ✓Ap

(b): The coenzyme supplements will be effective; ✓El The coenzyme supplement supports aerobic respiration and ATP production, which may improve energy availability and partially reduce fatigue however, it does not; correct the CFTR mutation, restore chloride transport, or prevent thick mucus forming, ✓Ee therefore continue with this strategy but only use as supportive treatment and not a long-term solution; ✓Ep

The gene therapy will be effective; ✓El Gene therapy introduces a functional CFTR gene so functional proteins are produced. This restores chloride transport, allowing water into the mucus thinning it, this addresses the root cause; although it risks an immune response as the immune system may attack the viral vector delivering the gene/may not reach all cells/treatment may need repeating; ✓Ee continue with this strategy but repeat it several times to ensure that it reaches many targeted cells; ✓Ep

The will Bronchodilators be effective; ✓El Bronchodilators relax bronchiole smooth muscle to widen the airways, giving rapid relief of wheezing, but they do not remove mucus or correct the defect and hence their effect is temporary; ✓Ee therefore continue with this strategy but only to roved short-term symptomatic relief since John's CFTR gene contains a base substitution (CTT to TTT) a mutation, which alters the amino acid sequence and tertiary structure of the CFTR protein, producing a non-functional protein that affects chloride ion movement; ✓Ep

1.6(a)

1.7(a) There was substitution gene mutation in the TP53 where T replaced C in the 3rd codon; ✓Al This was transcribed into a changed codon on mRNA which attracted a changed anticodon on tRNA carrying a different amino acid changing sequence of amino acids in TP53 protein; ✓Ae This faulty protein failed to monitor DNA for damage, repair and trigger apoptosis leading to limited control of the cell cycle leading to very high division rate causing cancer; ✓Al

Tumour size was larger in Mary; ✓Al because of high rate of abnormal cell division rate; ✓Ae due to faulty protein, cell cycle was not regulated and cells divided abnormally and at a large rate resulting into a group of large abnormal cells causing large tumour causing a painless lump in the breasts the many rapidly dividing rates consumed a lot of ATP reducing ATP available for muscle contraction resulting into fatigue. Cells also oxidized many stored organic compounds lie fats and causing weight loss; ✓Ap

Average number of mitochondria per cell in Mary was higher; ✓Al because of large tumour / high division rate; ✓Ae Tumour cells divide rapidly requiring large amount of ATP for DNA replication, protein synthesis spindle formation and cytokinesis stimulating synthesis of many mitochondria per cell; ✓Ap

ATP produce per glucose molecule was slightly higher in Mary; ✓Al because of high division rate; ✓Ae This required more ATP made mitochondria to harvest as much ATP from each glucose molecule as possible this required more oxygen consumption lowering oxygen in blood causing shortness of breath ; ✓Ap

(b) Gene therapy will be effective; ✓El because it will restore the normal / functional TP53 gene into tumour cells via a viral vector leading to transcription into a correct TP53 protein that will monitor DNA and cause repair and control cell cycle addressing the root cause However the viral vector may not reach in all cells / immune system may attack and destroy the viral vector; ✓Ee Therefore continue with this strategy but use repeated targets to ensure delivery into most targeted cell; ✓Ep

Chemotherapy combined with drugs that reduce glucose uptake by tumor cells will be effective; ✓El because chemotherapy drugs contain antimetabolites that block DNA replication and spindle formation killing rapidly dividing cells and drugs that

reduce glucose entry into tumor cells will limit glucose reducing ATP available to power rapid cell division of tumour cells However chemotherapy is nonspecific and may also kill normal rapidly dividing cells e.g in the bone marrow and gut epithelium causing anaemia; ✓Ee Therefore continue with this strategy but use targeted chemotherapy that only attacks breast tumour cells and give glucose uptake inhibitors in low doses to avoid very low glucose uptake by normal cells; ✓Ep

1.8(a) Mukasa had a gene substitution mutation at position 342 of DNA sense strand;

✓Al This code was transcribed into a changed codon in mRNA which was attached to ribosome and attracted a different anticodon on tRNA carrying a changed amino acid than the normal one this changed amino acid sequence in the antitrypsin protein formed; ✓Ae since the protein had a different sequence it did not fold properly to inhibit neutrophil elastase. Therefore this enzyme elastase was not inhibited and kept digested elastin like in the lungs preventing normal elastic recoil during expiration, lungs remained inflated causing hyperinflation and making the chest to be barrel shaped; ✓Ap

Blood serum AAT concentration was lower than normal in Mukasa; ✓Al due to DNA mutation; ✓Ae which could not code for proper arrangement of AAT protein during transcription and translation causing low manufacture of AAT in blood; ✓Al

Mukasa had low mitochondrial oxygen consumption than normal; ✓Al because lungs remaining ventilated inflated / low elastin in alveoli; ✓Ae this limited normal elastic recoil to cause efficient expiration and some air remained in lungs for long time reducing oxygen diffusion gradient, low oxygen in blood and delivery to the tissues reduced mitochondrial oxygen consumption. This low oxygen entry into blood caused shortness of breath and intolerance to exercise; ✓Ap

Relative mass of elastic tissue in alveoli in Mukasa was lower than normal; ✓Al due low inhibition of elastase by AAT; ✓Ae Since Mukasa had lower AAT, Elastase was produced by white blood cells and digested much elastin reducing its mass in alveoli; ✓Ap

(b) Direct oxygen therapy for both patient would effective; ✓El because Supplemental oxygen increases the partial pressure of oxygen in the alveoli, increasing the diffusion gradient for oxygen into blood. This temporarily improves

tissue oxygen delivery and aerobic respiration but Oxygen therapy does not stop destruction of alveolar elastin, does not restore damaged alveolar walls; ✓Ee therefore Oxygen therapy should be used temporarily mainly to relieve symptoms but should not be considered the definitive treatment for either patient; ✓Ep

Intravenous α1-antitrypsin (AAT) augmentation therapy for Mukasa would Work; ✓El because AAT augmentation therapy increases concentrations of functional α1antitrypsin which inhibits neutrophil elastase within the lungs, thereby reducing further degradation of alveolar elastin. but The treatment cannot regenerate elastin that have already been destroyed hence its ineffective if lung damage has already become severe; ✓Ee therefore AAT augmentation therapy should be started immediately to increase AAT levels in Mukasa but should be a life long treatment since Mukasa's condition is genetic; ✓Ep

Antioxidant therapy together with smoking cessation for Mutabazi would be good; ✓El Smoking cessation removes the major source of oxidants and toxins responsible for airway irritation. Antioxidants neutralize reactive oxygen species generated by cigarette smoke, reducing oxidative stress and helping preserve α1antitrypsin activity. but Antioxidants alone cannot reverse existing destruction of alveoli or restore lost lung tissue and Cessation of smoking requires cooperation from Mutabazi; ✓Ee therefore Mutabazi should immediately stop smoking antioxidants and also treatment to restore damaged lung tissue; ✓Ep 1.9(a)

The number of chromosomes in giant apple cells was twice the normal number; ✓Al because of agrochemical destroying Microtubule formation; ✓Ae This resulted into failure to synthesize spindle-fibres disrupted separation of chromosomes during Metaphase 1 of Mitosis making all to go the single cell doubling the chromosome number this resulted to larger cells and large fruits than normal; ✓Ap

Average surface area to volume ratio of cells was lower in Giant apple cells; ✓Al because of higher cell volume; ✓Ae when volume increases the surface area to volume ratio decreases since when size increases volume increases at a much faster rate than surface area does; ✓Ap

Ethanol concentration was highest in Deep core of treated apple; ✓Al due to high rate of anaerobic high rate of anaerobic respiration; ✓Ae being large in size the deep core is far away from the surface making diffusion of oxygen inefficient, as a result of high ATP demand due to a large cell volume the cells engage in anaerobic respiration generating huge amounts of Ethanol. Ethanol in large quantities s toxic

to cells resulted to death of cells in the deep core causing rad rotting of core of the apples; ✓Ap

Ethanol concentration in outer skin was slightly higher than normal; ✓Al due to being near the surface where there is faster diffusion of oxygen due to shorter distance from the surface; ✓Ae Therefore the cell mainly generated ATP needed through aerobic respiration, however due to high ATP demand as a result of large cell volume the cells still engage in anaerobic respiration generating Some ATP the ethanol was not to much and the cells could tolerate it so they remained healthy; ✓Ap

ATP yield per glucose molecule was higher in normal apple and outer skin of giant apple but lower in inner core of Giant apple; ✓Al because of large diffusion distance for oxygen; ✓Ae Small apples are small in size so diffusion of oxygen efficient over entire apple and skin cells of giant apple are near the surface for efficient oxygen diffusion however deep core can easily get oxygen by diffusion and engage mainly in anaerobic respiration product which yields only 2 ATPs per glucose molecule;

✓Ap

Duration during S phase of cell cycle was longer in Treated apples than normal; ✓Al due to large volume;✓Ae and hence much more materials need to be

synthesized for two large daughter cells which requires time; ✓Ap P yield per glucose molecule was higher in normal apple and outer skin of giant apple but lower in inner core of giant apple; ✓Al because of short diffusion distance diffusion distance for oxygen; ✓Ae shorter diffusion distance led to effcent dffuson of oxygen to the cells needed anaerobic respiration to generate ATP mang the cells n small apples and outer skin of large apples to make more at per mitochondria.
(b)

Spraying with chemicals that increase with chemicals that increase permeability of cell walls and cell membranes to oxygen will be effective; ✓El because it allows entry of oxygen easily reducing anaerobic respiration in the deep core avoid Toxicity of ethanol and rotting. However it makes cell walls to become leaky allowing water loss / allows entry of Pathogen infection/;✓Ee Therefore Continue with this strategy but use low optimal stable concentration with short exposure time; ✓Ep

Abandoning giant apples and carrying out selective breeding will be effective; ✓El because selective breeding maintain Normal number of chromosomes and cell volume allowing efficient gaseous exchange and no genetic instability of decreased

rotting in fruits However Standard apples still remain small and cannot give high yield as giant apples and this process is very slow; ✓Ee Therefore Continue with this strategy but control pollination cross pollination and use vegetative propagation while planting to maintain high yielding apples; ✓Ep

ITEM 2.1

Photosynthetic rate in maize was highest in green houses D and B; ✓Al because of temperature of 32°C which is near optimum for their CO2 fixing enzyme;✓Ae maize use PEPCO to fix CO2 and works best at 35°C. a temperature of 32°C in these greenhouses B and D is near this optimum and fix CO2 which rapidly fix CO2 which increases the rate of photosynthesis / productivity; ✓Ap

Photosynthetic rate in beans was highest in green house A and C at 22°C irrespective of hormone type; ✓Al because the temperature was near optimum for

CO2 fixing enzymes in beans; ✓Ae Beans use RuBP Carboxylase which works optimally at 25°C So a temperature of 22°C in green house A and C provided Stimulated products Carbon dioxide fixation increasing rate of photosynthesis in Beans; ✓Ap

Percentage germination in maize was highest in green house D; ✓Al because of being at a temperature of 32°C and having gibberellins; ✓Ae temperature of 32°C provided a conditions for germinating enzymes in maize whereas the gibberellin stimulated the synthesis of α-amylase that in the aleurone layer that hydrolyses stored food and promote germination; ✓Ap

Percentage germination in beans was highest in greenhouse A; ✓Al because of 22°C temperature near optimum of germinating enzymes in beans and presence of gibberellins; ✓Ae The temperature stimulated for working of germinating enzymes in Beans while the gibberellin stimulated synthesis of α-amylase that hydrolyses stored food in the cotyledons leading to high percentage germination; ✓Ap

(b) Plant Maize in green house D at 32°C to ensure high rate of photosynthesis;

✓El since this temperature is near optimum temperature for PEPCO enzyme in maize this will result into rapid fixation of CO2 and high rate of photosynthesis producing more organic products. However greenhouse block, wind currents of

limiting pollination of maize flowers; ✓Ee therefore continue with this method but install circulation fans to create air flow during flowering time; ✓Ep

Plant beans in green house A at 22°C to ensure high rate of photosynthesis; ✓El This temperature is near optimum for RUBISCO enzyme which will ensure high rate of CO2 fixation and rapid photosynthesis increasing productivity however there will be limited pollination due to greenhouses limiting wind movements and blocking pollinating insects; ✓Ee therefore continue with this strategy but Introduce pollinating insects during a period of flowering to ensure high pollination; ✓Ep

Provide gibberellin in green houses of both maize and beans during germination to increase percentage germination; ✓El because gibberellin stimulate synthesis of αamylase which catalyses hydrolysis of stored food needed for the embryo during germination however excess gibberellins during early stages may result into etiolation of seedlings and lodging; ✓Ee therefore continue with this method but apply only the small / recommended quantity; ✓Ep

ITEM 2.2:

(a) The % flowering in chrysanthemum plant was highest in green house A; ✓Al because it had the longest uninterrupted dark period of 14 hrs; ✓Ae Since they were short day plants, Long dark periods longer than a critical period make most of the phytochrome to be in the Pr form which n short-day plants stimulates production of florigen hormone that causes flowering in short day plants; ✓Ap

The mean shoot length in chrysanthemum plants was longest in green house D; ✓Al because it had the shortest uninterrupted dark period; ✓Ae this remained shorter than the critical and most phytochrome remained in Pfr form. This inhibit flowering in short day plant and they continued to grow vegetatively increasing shoot length to the longest height; ✓Ap

The rate of branching was higher in plants whose apices were removed compared to those whose shoot apices remained intact; ✓Al because Removal of shoot apex removes apical dominance; ✓Ae Shoot apex produce auxins which flow downward and inhibit lateral bud development. If the tip is removed this inhibition is removed and lateral bud develop and become side branches whch caused many branches in plants whose shoot was removed; ✓Ap

(b) Plant Chrysanthemums in green house A where there is 10 hours light and 14 hours darkness; ✓El This will provide enough uninterrupted dark period greater than critical night length making their phytochromes to exist mainly in Pr form which stimulates flowering hence many flowers for ornamental use However low light period will reduce photosynthesis to during yield early growth limiting flower yield; ✓EeTherefore continue with this strategy but only provide a long night and a short day during time for flowering; ✓Ep

plant spinach in green house D; ✓El This is longer day than critical which will make most of their phytochromes to appear in phytochrome red far red form to stimulate flowering However the vegetative growth will be low for consuming; ✓Ee therefore proceed with this strategy but provide this condition at flowering stage to harvest both vegetables and flowers; ✓Ep

Remove apices in both plants to stimulate high branching; ✓El because removing the tip removes auxin source removing inhibition on lateral buds stimulating many branches on which flowers will form however wound caused may become entry points for fungi and bacteria causing rotting; ✓Ee Therefore continue with this strategy but do so with a sterile blade; ✓Ep

2.4 (a) Germination is highest in Greenhouse 4 (25°C with gibberellins); ✓Al because germination enzymes work close to their optimum at 25°C, whereas at 32°C enzyme activity is reduced by partial denaturation. And presence of gibberellins; ✓Ae This is because I"n greenhouse 4, gibberellins stimulate amylase, which hydrolyses stored starch into sugars for respiration and embryo growth, so the combination of optimum temperature and gibberellins gives the highest germination rate as in Greenhouse 4; ✓Ap

Photosynthetic rate is high at 32°C and low at 25°C regardless of hormone type; ✓Al because enzymes such as RuBisCO are more active at 32°C, increasing carbon fixation and glucose production; ✓Ae temperature is therefore the determining factor, while the growth regulators have little influence on photosynthesis; ✓Ap

Gibberellin-treated plants produce the most branches (highest in Greenhouse 2 with gibberellins+ 32°C); ✓Al because of gibberellins and a higher temperature;

✓Ae gibberellins promote lateral bud growth while auxins maintain apical dominance, and higher temperature increases the rate of these enzyme-controlled growth processes; ✓Ap

(b) Use conditions of Greenhouse 4 during germination; ✓El Greenhouse 4 (25°C with gibberellins) gives the highest germination rate hence most seedlings from seeds, but photosynthetic rate is low at 25°C it is unsuitable for later growth and fruiting; ✓Ep therefore use it for the germination stage only; ✓Ep

Use to conditions of Greenhouse 2 for growth and branching; ✓El Greenhouse 2 (32°C with gibberellins) gives the highest photosynthetic rate and most branches, providing more fruiting sites, but its lower germination rate; ✓Ee therefore use it only after seedlings are established; ✓Ep

ITEM 2.5

(a) Photosynthetic rate in beans was highest in greenhouse 2 (20°C + High CO2); ✓Al because of temperature near optimum for their photosynthetic enzyme and High CO2; ✓Ae Beans are C3 plants and use RuBP carboxylase which has an optimum temperature of 25°C a temperature of 20°C was near optimum so the enzyme fixed near optimum level also it has a low affinity for CO2 so high CO2 can easily outcompete oxygen resulting into low photorespiration and high rate of CO2 fixation; ✓Ap

Rate of photosynthesis in maize was highest in greenhouse 3 and 4 minimal 35°C despite the level of CO2; ✓Al because 35°C is a temperature optimum temperature for their CO2 fixing enzyme; ✓AeC4 maize uses PEPCO which has an optimum temperature of 35°C and has a high affinity for CO2 hence can fix CO2 at a very high rate in both ambient and high level to ensure rapid rate of photosynthesis; ✓Ap

The water use efficiency in beans was highest in greenhouse 2; ✓Al because the stomata was partly open due to high CO2 in this green house; ✓Ae beans would partly close the stomata and still get enough CO2 for photosynthesis this reduced the rate of stomata transpiration and reduced water loss. Although in 4 stomata were also partly open, the high temperature increased photos the rate of evaporation in the partly open stomata; ✓Ap

Water use efficiency in maize highest in greenhouse 3; ✓Al because the stomata opening was slightly reduced; ✓Ae since maize use PEPCO with an optimum temperature of 35°C and has very high affinity for CO2 so even in ambient CO2 it can fix CO2 well and with lowest water loss; ✓Ap

Biomass accumulation in beans was lowest in greenhouse 3; ✓Al because the temperature of 35°C is far beyond the optimum for Rubisco and ambient CO2; ✓Ae high temperature beyond optimum denatured many CO2 fixing enzyme and promoted high rate of photorespiration and ambient CO2 promoted photorespiration since oxygen could easily out compete CO2 for active sites of Rubisco; ✓Ap

Biomass accumulation was higher in greenhouse 3 and 4 at 35°C despite the level of CO2; ✓Al because they 35°C is optimum for PEPCO; ✓Ae Pepco has a high affinity for CO2 and can fix CO2 well even in ambient and high conditions giving high rate of photosynthesis and accumulation of biomass; ✓Ap

(b) Maize should be planted in greenhouse 3 conditions; ✓El because it provides high rate of photosynthesis in maize with high water efficiency and accumulate high biomass and yield however greenhouse can limit wind movements and pollination during flowering; ✓Ee therefore continue with this method but enhance air movements to pollinate using moving wind using electric fans; ✓Ep

plant beans in greenhouse 2 conditions; ✓El Since low temperature of 20°C is near optimum for Rubisco and high CO2 reduces photorespiration by out competing oxygen to active sites of Rubisco however greenhouse limits entry of pollinating insects to pollinate the flowers causing poor yields; ✓Ee therefore Continue with this method but put pollinating insects like bees in the greenhouse during flowering period; ✓Ep

Irrigate the beans to ensure that there is enough water in the soil; ✓El High water content in the soil will eliminate conditions of water stress allowing beans to keep their stomata wide open allowing entry of CO2 for photosynthesis however excessive irrigation may result into leaching of nutrients and soil erosion; ✓Ee Therefore Continue with this method but ensure only adequate amount of irrigation required to avoid leaching;✓Ep

ITEM 2.6

(a) Percentage germination of both millet and Beans were lowest in Cycle 3 compared to cycle 2 and cycle 1; ✓Al because seeds planted from cycle 2 had high level of abscisic acid; ✓Ae Abscisic acid in seeds promote seed dormancy by inhibiting production of Gibberellins which are need to stimulate synthesis of

αamylase to hydrolyse stored food during germination so seeds failed to germinate; ✓Ap

The mean rate of photosynthesis in Beans was lowest in Cycle 3; ✓Al because Irrigation was reduced and stopped completely in cycle 3; ✓Ae This reduced amount of water in the soil and created water stress but since Maize can cause accumulate CO2 and close the stomata it is good at efficiently use of water but for C3 plants they closed their stomata to control water loss and have Kranz anatomy so entry of CO2 was reduced creating lowest rate of photosynthesis; ✓Ap

In all 3 Cycles rate of photosynthesis in Millet was higher than that of C3 plants; ✓Al because Millet being a C4 uses PEPCO which has a higher affinity for CO2; ✓Ae C4 use PEPCO with a higher affinity for CO2 hence can rapidly fix carbon dioxde increasing the rate of photosynthesis C3 (beans) use Rubisco which has a lower affinity for CO2 and can easily undergo photorespiration hence lower photosynthesis rate; ✓Ap

percentage of flowering of Maize was lowest in cycle 2 and that of beans remained high through; ✓Al because the black curtain failed in cycle 2 disrupting the uninterrupted dark period; ✓Ae since millet is a short long day plant lack of a dark period longer than critical period resulted in phytochrome being in Pfr form which in short day plant inhibit conversion of a precursor to florigen hormone to bring about flowering. In beans flowering was not affected because they are day; ✓Ap

Seed yield in millet was lowest in cycle 2; ✓Al due to interruption of dark period low percentage flowering; ✓Ae because millet is a C4 plant and disruption of the dark period made their phytochrome to appear in Pfr form which inhibits flowering in short day plants therefore no face few fruits were formed from few flowers formed hence low rate of seed formation; ✓Ap

Seed yield in Beans were lowest in Cycle 3; ✓Al due to closure of irrigation; ✓Ae This reduced water in the soil and created water stress making C3 plants to close their stomata preventing entry of CO2 hence resulting into low rate of photosynthesis and low manufacture of food to store in the seeds; ✓Ap

(b) Millet should be provided with long periods of the dark longer than the critical night period; ✓Ep This will stimulate flowering since most of the phytochromes will appear in Pr form and stimulate a precursor conversion to florigen hormone making

them to flower highly and produce yield however in case of openings in the curtain, brings flashes of light and drastically reduce flowering; ✓El curtain should be inspected and ensure there is no leakage during the flower period;✓Ep

Beans should be irrigated always and put at a lower temperature of 35°C; ✓El beans are C3 plants and use Rubisco enzyme to fix CO2 which works best around 25°C and irrigation ensures availability of water avoiding water stress so stomata keep open and allow in CO2 for high photosynthesis however excessive irrigation may result to leaching of nutrients; ✓Ee Therefore continue with this method but only irrigate with adequate amount of water to avoid leaching; ✓Ep

Treat the seeds of both millet and beans with Gibberellins before planting; ✓El this will break dormancy since gibberellins stimulate synthesis of α-amylase that hydrolyses stored food for the embryo however Gibberellins may make the seeds to germinate during storage; ✓Ee therefore only treat seeds with Gibberellins during germination time; ✓Ep

ITEM 2.7

(a) The Tea Net Assimilation Rate decreased from 14.2 µmol CO_2 m^{-2} s^{-1} in Plot A (open sunlight) to -1.5µmol CO_2 m^{-2} s^{-1} in Plot B; ✓Al (70% shade by Napier grass). The dense canopy formed by the overgrown Napier grass reduced the amount of light reaching the tea bushes; ✓Ee This caused the rate of photosynthesis to below the rate of respiration. The tea bushes therefore consumed more organic matter through respiration than they produced by photosynthesis, resulting in a negative net assimilation rate and reduced biomass Reduced photosynthesis lowered carbohydrate production and chlorophyll synthesis. Chlorophyll molecules gradually degraded faster than they were synthesized, causing chlorosis (pale green leaves) and further reducing the photosynthetic rate; ✓Ap

The Tea Stem Elongation Rate increased from 12.0 mm/week in Plot A to 45.0 mm/week in Plot B; ✓Al due to Shading; ✓Ae alters the red to far-red light ratio detected by phytochrome pigments. This stimulates increased synthesis of auxins which promote rapid cell elongation within stem internodes; AC Increased auxin production at the shoot apex caused apical dominance. Auxins moved downward from the apical bud and inhibited the growth of lateral buds. Hence, the Average Number of Lateral Bud Sprouts decreased from 22 in Plot A to 2 in Plot B; ✓Ap

The farmer also observed that the soil beneath the dense grass dried rapidly during brief sunny intervals; ✓Al The overgrown Napier grass had an extensive fibrous

root system and a high transpiration rate; ✓Ae It rapidly absorbed soil water before it could be utilized by the tea bushes; ✓Ap

In Plot C, Napier grass was regularly cut and an auxin blocker was sprayed on tea bushes. Cutting grass reduces shading and allows more light to reach the tea bushes; ✓Al and increase photosynthesis; ✓Ae hence tea Net Assimilation Rate increased to 13.8 µmol CO_2 m^{-2} s^{-1}; ✓Ap

Blocking auxin action reduced apical dominance; ✓Al allowing lateral buds to develop into numerous branches; ✓Ae suitable for harvesting hence the stem elongation rate fell to 4.0 mm/week, while lateral bud sprouts increased to 48 per bush; ✓Ap

In Plot D, exogenous auxins applied stimulate stem elongation; ✓Al while suppressing lateral bud growth through apical dominance; ✓Ae hence the stem elongation rate increased to 52.0 mm/week, while lateral bud sprouts declined to only 1 per bush; ✓Ap

(b) Regularly slashing the Napier grass to less than 0.5 metres is good; ✓El Reduces height of Napier grass and allows more sunlight to reach the tea bushes, increasing the rate of photosynthesis / Also lowers auxin production induced by shade, thereby reducing excessive stem elongation. Less competition for soil water; improves water availability to tea bushes. But if the grass is cut too frequently, its effectiveness in controlling soil erosion may decrease/ Regular slashing requires continuous labour and increases costs; ✓Ee Therefore This should be practiced as long term strategy as maintaining the grass below 0.5 metres balances control of soil erosion with adequate light penetration for tea production / Slashing be done at appropriate time intervals; ✓Ep

Mechanically clipping the apical buds of the tea bushes would work; ✓Ep Removal of the apical bud eliminates the source of auxins responsible for apical dominance and allows development of numerous side branches, increasing the number of pluckable shoots. however, frequent clipping requires skilled labour and increases management costs/Excessive pruning may temporarily reduce leaf area available for photosynthesis/Fresh pruning wounds may increase susceptibility to pest and pathogen attack; ✓Ee Therefor: Apical buds should be clipped only at appropriate time intervals; ✓Ep

Applying a light organic mulch around the tea roots is good; ✓Ep Organic mulch reduces evaporation from the soil surface, conserving moisture around the root system / It moderates soil temperature / suppresses weed growth/ gradually releases nutrients as it decomposes, improving soil fertility; But Organic mulch may retain too much moisture, encouraging fungal diseases and rotting of roots/ may introduce pests and weeds; ✓Ee Therefor Light organic mulch should be applied but together with other methods intended to reduce shading by Napier grass; ✓Ep

ITEM 2.8.

(a) At 20°C, Net Photosynthesis of Species A was higher than net photosynthesis of Species B; ✓Al because Species A is a C3 plant and uses RuBP Carboxylase whose optimum is about 25 and a temp of 20 is near the optimum for their CO2 fixing enzyme; ✓Ae these plants rapidly fix CO2 and have high rates of photosynthesis. Species B use PEPCO whose optimum is high (35°C) So at a temperature of 20°C, the enzyme was inactive hence lower rate of photosynthesis; ✓Ap

At 35°C Net photosynthesis was higher in Species B; ✓Al because the temperature was near optimum for the carbon dioxide fixing enzyme of Species B; ✓Ae Species B is a C4 plants which use PEPCO enzyme. This enzyme works optimally at about 35°C hence those plants rapidly fixed CO2 and produced high rates of photosynthesis. ✓Ap

At both temperatures photorespiration was higher in Species A than B; ✓Al because Species A are C3 plants that use Rubisco to fix CO2; ✓Ae this enzyme has affinity for both oxygen and CO2 hence can sometimes combine with Oxygen to produce Non carbohydrate compounds. ✓Ap

b) Grow species A in a green house with enriched CO2; ✓El This will reduce photorespiration since CO2 will easily outcompete oxygen for the Active site of Rubisco and increase the efficiency of photosynthesis and productivity; However green house traps humid stagnant air creating breeding ground for fungal pathogens; ✓Ee therefore continue with this method but spray always with fungicides to prevent growth of such fungi; ✓Ep

Irrigate species A with regularly; ✓El this creates enough Soil moist and reduces closure of stomata due to water stress allowing stoma to remain open and allow in

CO_2 for high rates of photosynthesis. However excessive irrigation may result to leaching of nutrients; ✓Ee therefore Continue with this strategy but irrigate with adequate amount of water to avoid leaching and soil erosion of nutrients; ✓Ep

Plant Species A and Species B on separate fields; ✓El because this reduces competition and since each specie requires different conditions these can be easily provided to each crop Animals will not easily feed on both legumes and grasses for balanced dietary requirement; ✓Ee Therefore continue with this strategy but cut and mix the two crops before giving them to animals; ✓Ep

ITEM 2.9

(a) Percentage germination of moringa seeds was higher during cool, moist conditions; ✓Al because of higher amount of Gibberellins than abscisic acid; ✓Ae
In moist conditions the seed imbibes water that stimulates synthesis of Gibberellins which stimulate synthesis of α-amylase in aleurone layer that hydrolyses food store into simple organic substances for use by the growing embryo; ✓Ap

Soluble sugars produced was higher in the cover crop than in moringa tree; ✓Al because Activity of PEPCO and Rubisco was higher; ✓Ae Cover crop being a C4 plant uses PEPCO with a higher affinity for CO_2 and its RuBP Carboxylase has a higher activity because C4 Concentrate CO_2 in form of malate and shield Bundle sheath cells from atmospheric oxygen therefore there is higher around these cells and no oxygen therefore there was more CO_2 fixation for use in photosynthesis to produce much more sugar; ✓Ap
Amylase activity (%) was higher during cool moist conditions than in hot, dry conditions; ✓Al Because of higher proportion of Gibberellin to Abscisic acid; ✓Ae High Gibberellins diffuse into aleurone layer of seed and stimulating production of enzymes like amylase. This high concentration of amylase bring about high activity during this condition; ✓Ap

The number of branches per tree of moringa were higher in presence of Cytokinins; ✓Al because of spraying with Cytokinins higher ratio of Cytokinins; ✓Ae Cytokinins are antagonistic to auxins which inhibit bud development but Cytokinins break bud dormancy and promote development of lateral buds to form many side branches; ✓Ap

(b) Treatment of all stored seeds with gibberellins immediately after harvesting will not be effective; ✓El because Gibberellins will break the natural dormancy and force seeds to start germinating prematurely which ruins storage life and causes abnormal growth however it can allow complete bypassing of natural seed dormancy allowing immediate planting and year round breeding cycles; ✓Ep Therefore abandon treating seed with gibberellins immediately after harvesting but treat them with abscisic acid to promote dormancy and safe storage and avoid germinating in stores; ✓Ep

Planting Moringa in green house with enriched CO2 will be effective at improving synthesis of medicinal products; ✓El Since Moringa is a C3 plant Enriched CO2 will outcompete oxygen for RuBisCO active site allowing faster fixation of CO2 and low respiration ensuring high yield however Green house may trap pests and fungi resulting in increased pest damage and fungal diseases; ✓Ee therefore Continue with this method but always spray pests using pesticides and fungi using fungicides; ✓Ep

Relocating the moringa tree plantation to a highland area will be effective at increasing synthesis of medicinal products; ✓El because at high altitude there is reduced partial pressure of oxygen reducing the rate of photo respiration in C3 moringa and increasing efficiency of photosynthesis However there is rampant soil erosion at high altitude with wastes away top soil containing nutrients if the highland is at very high altitude growth may be inhibited by very low temperature; ✓Ee therefore continue with this strategy but plant them at optimal highland zone; ✓Ep

Item 3.1

(a) Blood flow to the skeletal muscles was highest during escape; ✓Al This is due to high levels of Adrenaline levels and highest Sino Atrial Node firing frequency; ✓Ae Sight of a wild animal stimulates secretion of high amount of Adrenaline hormone that stimulates vasodilation in skeletal muscles and also stimulates heart rate increase. Actively respiring muscle cells also produce large quantities of CO2 which cause local vasodilation resulting into increased blood flow to the skeletal muscles required for taking the required glucose and oxygen; ✓Ap

Blood flow to the intestines digestive organs decreased during the race; ✓Al due to adrenaline levels being high; ✓Ae High levels of adrenaline in blood and CO2 from most active muscles stimulate vasoconstriction in other non-vital body organs like

the digestive organs to ensure high blood flow to the active muscles during response; ✓Ap

Sinoatrial Node firing frequency was highest during the escape; ✓Al because high levels of adrenaline in blood and high activity of the muscles during escape; ✓Ae Increase of CO2 in blood is detected by chemoreceptors in aorta and carotid bodies which send impulses to the vasomotor center cardioacceleratory centre which in turn sends impulse via sympathetic nerves to the SAN to increase Activity of SAN and heart rate; ✓Ap

During escape blood Adrenaline concentration increased; ✓Al because of sight of a wild animal charging at him acted as fear stimulus; ✓Ae This stimulated the cerebral cortex and the hypothalamus resulting into activation of sympathetic nervous system which in turn stimulates the adrenal medulla to secrete large quantities of adrenaline to stimulate the body for fight / flight response; ✓Ap

Membrane potential of local thigh neurones was very low Repolarised;✓Al Due to the injection of the Lignocain drug that blocks Na+ Voltage gated channels; ✓Ae This prevented entry of Na+ into the nerve cell to cause depolarisation and the sodium potassium pump pumped out Na+ out of the axoplasm resulting into 70mV or resting potential making the nerves unable to transmit pain information to the brain hence the area becoming numb and insensitive to pain and touch; ✓Ap

After the escape Adrenaline blood flow to the intestines skeletal muscles and sinoatrial node firing frequency decrease while blood flow to the digestive organs increased; ✓Al Because the fisherman reached safety in the cave; ✓Ae which stimulated parasympathetic nervous system to inhibit secretion of adrenaline, send impulses via vagus nerve to slow down heart rate to normal and dilate blood vessel in digestive system to allow digestion to continue; ✓Ap

(b) Administration of the medicine was effective; ✓El because the medicine blocked gated channels and prevented generation and transmission of impulses from the pain receptors to the brain which removed a sensation of pain making it easier to clean and suture the wound however the local drug did not reduce bleeding and could not prevent entry of microorganisms into the open wound; ✓Ee therefore continue with this method to cover the wound to prevent excess bleeding and administer antibiotics to the wounded tissue to prevent entry of bacteria; ✓Ep

Apply broad spectrum antibiotics; ✓El this will inhibit entry and growth of the microorganisms like bacteria into the body through the wound and promote healing however some bacteria are resistant to some antibiotics; ✓Ee therefore continue with this strategy but choose the ideal antibiotic or apply a different type of antibiotic in case of resistance; ✓Ep

Do blood transfusion in case the blood lost is too much; ✓El This will increase blood in the body immediately thus balancing pressure and supply of nutrients reducing low blood pressure but if the blood is not examined at blood group compatibility and Rhesus factor it may cause agglutination and death; ✓Ee Therefore do grouping and cross matching before transfusion; ✓Ep

3.2 (a)

Tail flicking was highest on day 3; ✓Al because of pain from biting insects; ✓Ae Tsetse flies possess biting mouth parts that penetrate the skin and stimulate pain receptors during in the skin to send impulses to the brain of cows which stimulates innate reflex to flip the tail and twitch the skin to get rid of such insects; ✓Ap

Tail flicking decreased from day 3 to day 10 of post infection; ✓Al because of high trypanosomal parasites in blood; ✓Ae thus high antigen density stimulated release of cytokines stimulate a feeling of fatigue to redirect energy from muscle movement to manufacture of antibodies to fight the pathogen; ✓Ap

Core body temperature increased and was highest during the day 10 of post infection; ✓Al because of highest trypanosome density in blood; ✓Ae The trypanosomes or White blood cells (neutrophils) trying to engulf them produce substances known as pyrogens which change the thermostat set point in the hypothalamus and set it to a higher temperature to stimulate defence response and fight the pathogen; ✓Ap

Daily pasture intake decreased thought from preinfestation to day 10 after infection; ✓Al because of biting insects making animals to move up and down the pasture instead of feeding; ✓Ae therefore less time was used in feeding leading to decrease in pasture intake. During heavy infestation animal lost appetite to feed due to pyrogens reducing pasture intake even further; ✓Ap

Plasma specific antibody concentration was highest on Day 10 post infection; ✓Al to highest number of pathogens/ trypanosomes in blood; ✓Ae These pathogens stimulates proliferation of B cell with a matching receptor to proliferate and produce large quantities of antibodies; ✓Ap

Plasma ADH levels were highest during day 10 of post infection; ✓Al because of very high body temperature; ✓Ae High body temperature stimulates sweating in order to cool the body which results to loss of large amounts of water and makes blood concentrated; concentrated blood stimulates the hypothalamus to stimulate posterior pituitary to release high amounts of ADH into blood to increase water Reabsorption in the kidney; ✓Ap

(b) Spraying the cattle with a synthetic insecticide spray will be effective; ✓El at Since it kills adult tsetse flies reducing tsetse fly bites reducing pain irritation and stress allowing cattle to graze normally and produce milk however spraying will not kill trypanosoma parasites and already infected cattle remain ill and may die to heavy infestation; therefore continue spraying with the synthetic insecticide; ✓Ee but treat the infected animals with appropriate drugs that kill trypanosomes in blood; ✓Ep

Administering a high dose antipyretic drug will be effective; ✓El since anti pyretic drugs reduce the hypothalamic temperature set point allowing loss of heat and preventing irreversible brain damage due to high temperature however parasites may continue to multiply in blood stream since fever is part of body innate immune response to increase response to pathogens which may delay recovery; ✓Ee Therefore Continue with administering antipyretic drugs but combine it with administering of a drug that kills the trypanosomes in blood; ✓Ep

Doubling the herds water allocation will not be effective; ✓El because ill cattle may have low appetites and may fail to take the water provided however water intake helps to replace water lost by evaporation during cooling allowing the animals to easily regulate temperature; ✓Ee Therefore abandon this strategy but instead provide shelters with cool temperature to avoid heat exhaustion in infected cattle; ✓Ep

Item 3.3

(a)Dissolved oxygen in affected pond was lower than that of affected pond; ✓Al due dumping of untreated wastes from agro processing factory; ✓Ae Untreated waste water contains large amount of organic matter which is rapidly decomposed by aerobic bacteria. Aerobic bacteria use the available oxygen reducing its concentration; ✓Ap

Blood Oxygen saturation was lower in fish from affected ponds; ✓Al because of lower dissolved oxygen in the surrounding water; ✓Ae This low dissolved in the water This reduced concentration gradient across the gill surface reducing the rate of diffusion of oxygen into blood of fish; ✓Ap

Heart beats in (beats min) was higher in fish from affected ponds; ✓Al because of low low dissolved oxygen; ✓Ae This low oxygen reduced aerobic respiration and cells engage in anaerobic respiration generating large amount of lactic acid. This lowered pH stimulating chemoreceptors in the aortic bodies to send impulse to the cardiac acceleratory centre which sent in turn send impulses to the SAN to increase heart rate; ✓Ap

Response time to external stimuli was higher in fish of affected pond; ✓Al due to low oxygen dissolved in water; ✓Ae this low oxygen reduced the efficiency of aerobic respiration reducing amount of ATP in cells. Since generation of impulses requires ATP, the ATP generation stimulus generator potential formed slowly increasing response time; ✓Ap

White blood cell count was higher than normal fish from affected ponds; ✓Al because of exposure to pathogen / substances that stressed fish; ✓Ae This foreign bodies / antigens stimulates rapid multiplication of white blood cells to carry out phagocytosis of the foreign bodies; ✓Ap

(b) install water treatment plant to treat the water before discharge; ✓El This involves sedimentation and reduces the organic load which reduces BOD preventing anoxic conditions however this is slow and fish will keep dying; ✓Ee Continue with this strategy but aerate the water to to avoid reduce death of fish; ✓Ep

Install machine to aerate the water; ✓El this will increase dissolved oxygen unit in a short period of time however strong aeration causes much water turbulence stressing fish and increases energy expenditure for swimming using ATP that

should be used for growth and immunity; ✓Ee therefore continue with this strategy but aerate occasionally and for short periods of time; ✓Ep

plant buffer zone of papyrus; ✓El papyrus roots provide physical filtration, they trap suspended solids, absorb nitrates and heavy metals reducing Eutrophication However if buffer plants die and decompose in the water their own organic matter creates BOD and lowers the dissolved oxygen creating hypoxic conditions; ✓Ee therefore lant that are tolerant ; ✓Ep

ITEM 3.4

(a) Core blood temperature in group B was Higher than Group A; ✓Al because Inadequate drinking water during periods of high environmental temperature; ✓Ae this reduced rate of heat loss through panting and evaporation and heat accumulates in the body raising body temperature; ✓Ap

Plasma ADH concentration was lower than those that were well adopted; ✓Al This is due to high body temperature;✓Ae. This high temperature denatured enzymes causing failure of endocrine system to secrete ADH hence low; ✓Ap

Urine osmotic concentration was higher in well adapted birds than in Group B affected birds; ✓Al due to hgh levels of ADH n blood; ✓Ae In group A osmoreceptors in the hypothalamus detect high plasma osmolarity stimulating the posterior pituitary to release ADH. ADH increases reabsorption of water in kidney tubules and collecting ducts by increasing aquaporins so little water is lost making urine very concentrated. In group B inadequate drinking water and heat stress disrupt production of ADH due to endocrine system failure hence blood volume decreases reducing evaporative cooling so core temperature raises further to 43.3; ✓Ap

Antibody concentration after vaccination in group B was lower than normal well adapted birds of group A; ✓Ae because of high core body temperature; ✓Ae high core body temperature reduces protein synthesis so immunoglobulins are not made since it may result to denaturation of enzymes involved; ✓Ap

Disease incidence during observation was higher in Group B than Group A; ✓Al because of lower concentration of antibodies; ✓Ae This decreases the humoral response and birds can not fight infection so pathogens multiply rapidly and cause symptoms and egg production stops due to use of energy to fight the infections; ✓Ap

(b) provide adequate cool drinking water and electrolytes; ✓El water intake results in evaporative cooling during panting, restores blood volume so vasodilation and blood supply to the skin increases leading to dropping of core body temperature However if the water contain excess electrolytes without balance can increase plasma concentration disrupting enzyme action; ✓EeTherefore continue with this

"strategy but pro per mix it with correct amount of Electrolytes; ✓Ep

Provide well ventilated housing with shade; ✓El This allows air circulation and increases the rate of heat stress loss by convection by flowing air currents reducing heat stress. However too much ventilation can result to excessive heat loss during the night when heat temperature drops causing birds to lose a lot of energy in shivering and thermogenesis reducing energy for egg production; ✓Ee therefore continue with this method but reduce the ventilation at night and during cold weather; ✓Ep

Reducing over crowding of birds in the grea house; ✓El This will reduce contact transmission of pathogen reduces metabolic heat per square meter However if stocking density of birds is too scattered they can not huddle during cold periods to conserve heat leading to energy waste; ✓Ee therefore continue with this strategy but leave the for adequate number of birds per sq in the house; ✓Ep

ITEM 3.5

(a) Core temperature of Okello was slightly lower than normal; ✓Ap because of immersion in freezing water; ✓Ae This caused partial failure of the thermoregulation however the body was still able to compensate for heat loss through increase in metabolic rate, vasoconstriction of arterioles in the skin, decreased blood flow to the skin, decreased radiation convection and conduction, inhibition of sweating raising hair and increased shivering; ✓Ap

Core temperature of Kasasa was much lower than normal; ✓Al because of being emersed in freezing water for a longer period of time; ✓Ae This led to breakdown of homeostatic systems causing heat to be lost easily. Hypothalamus itself becomes cold and fails to send impulses; ✓Ap

Shivering rate was highest in Okello but absent in Kasasa because of shorter time of exposure to cold water; ✓Ae Okello was exposed for a short time therefore homeostatic system was still compensating for heat loss through shivering. In

Kasasa there was no shivering because the homeostatic systems had already failed to function due to hypothermia (Too cold for enzymes to hydrolyse ATP and); ✓Ap

Mean arterial blood pressure of Okello was higher than normal; ✓Al because of secretion of adrenaline hormone in blood; ✓Ae Cold causes sympathetic activation causing adrenaline secretion which causes constriction of capillaries in the skin and other extremities to reduce heat loss by conduction and convection; ✓Ap

Mean arterial pressure of very low in Kasasa; ✓Ap due to failure of circulatory collapse due to hypothermia; ✓Ae prolonged shock cause failure to vasoconstrict vessels dilate and blood pours into extremities and the nervous system fails to regulate pressure falling below normal; ✓Ap

Blood level of glucose was lowest in Kasasa; ✓Al because of exhaustion of glucose due to severe muscle usage; ✓Ae Kasasa spent much more time so most glucose was losing during shivering. Low glucose causes semi consciousness and prevents re warming; ✓Ap

Blood glucose level in Okello was lower than normal although higher than Kasasa; ✓Al because he had spent less time in cold water because of shivering; ✓Ae shivering uses glucose and glycogen stores to generate ATP and heat as being younger he had more muscle mass and could sustain shivering for long time due to high glycogen stored;✓Ap

(b) Intravenous injection of isotonic fluid into Kasasa was effective; ✓El because stays in extracellular it increases plasma volume increases plasma venous return increasing stroke volume and raising pressure back to normal. This improves blood flow to the brain reversing oxygen delivery to the brain for reversing semi consciousness. However if the fluid is not first warmed it may result to further drop in core temperature;✓Ee Therefore continue with this strategy but warm the injected fluid to 40°C before infusion; ✓Ep

Transferring him to a warmer room at 32°C will be effective; ✓El because heat will be absorbed into the body by convection and conduction and radiation Raising core body temperature, However if the temperature becomes too warm, heat receptors in the skin will cause vasodilation of skin capillaries further reducing blood pressure / does not correct low blood sugar; ✓Ee therefore continue with this

method but combine with giving him a glucose solution to increase blood glucose level; ✓Ep

Offering a strong sucrose solution immediately will not be effective; ✓El because sucrose is a table sugar made up of glucose and fructose it can only be broken down in the intestines where there is sucrase enzyme since there is no sucrase in blood it will not raise blood glucose level. However it is hypertonic and will cause outflow of water from tissues into blood by osmosis increasing blood volume; ✓Ee Therefore Abandon this method and instead use glucose which can enter directly into the cells; ✓Ep

ITEM 3.6

(a)The Erythrocyte cytoskeleton was rigid instead of being flexible; ✓Al because of low blood saturation with oxygen than normal; ✓Ae because of a mutation, at low oxygen saturation the sequence of amino acid is changed as a result of substitution gene mutation where Valine replaces Glutamic acid making the red Haemoglobin to become less soluble at low partial pressure of oxygen in blood. These crystallize and become stiff elongating the shape of RBC and making them rigid. Rigid RBC clogged capillaries in legs causing pain; ✓Ap
Blood saturation was low in Amina after the race;✓Al because of rigid Erythrocytes and crystallised haemoglobin which is unable to carry oxygen; ✓Ae resulting low oxygen carriage from the lungs giving Amina abnormally low oxygen saturation; ✓Ap

ATP in muscle cell in Amina was lower than normal; ✓Al because of poor oxygen supply to the muscle cells due to rigid Erythrocytes; ✓Ae This made the muscle cells to rely on anaerobic respiration to generate which generates low amount of ATP resulting into a feeling of fatigue; ✓Ap

Plasma pH in Amina was lower in Amina; ✓Al due to Erythrocyte being rigid which prevented blood flow; ✓Ae cell produced a lot of CO2 which was not easily transported back to lungs for exhalation and excess lactic acid from anaerobic respiration which reduced pH of the blood in Amina; ✓Ap

(b) High flow supplemental oxygen will be effective; ✓El because it will increase alveolar oxygen partial pressure and increase oxygen saturation of blood in Amina and prevent crystallisation of Haemoglobin improving oxygen supply to muscles

and improve ATP production. However too high flow can suppress respiratory drive and cause oxygen toxicity; ✓EeTherefore continue with this method but only use it for a short time to treat acute oxygen crisis; ✓Ep

Intravenous fluids with glucose will be effective✓El since these fluids reduce blood viscosity improving blood flow, whilst washing out lactic acid while delivering glucose to the tissues for use in ATP generation; However over hydration can cause fluid overload providing dangerous stress on circulatory system and organs; ✓Ee Therefore Continue with this method but controlled amount of fluid should be inserted during rise of symptoms; ✓Ep

Prohibiting Amina from engaging in more Athletics will be effective; ✓El since it prevents Amina from doing intense exercise which lower blood oxygen saturation and causes Haemoglobin to crystallize However sickle cells will occasionally form and block capillaries in the kidney and legs causing acute pain; ✓Ee Continue with this strategy undergo regular transfusion to provide healthy normal red blood cells; ✓Ep

ITEM 3.7

(a)ADH concentration in blood was higher in Healthy student than David; ✓Al because of consuming of large amounts of alcohol by David; ✓Ae Alcohol is a diuretic compound and when taken it was absorbed into blood stream where it travelled through blood to the hypothalamus inhibiting release of ADH by posterior pituitary; ✓Ap

Cardiac output was lower in David; ✓Al due to low dehydration; ✓Ae David had low ADH in blood therefore kidneys were not stimulated to reabsorb more water from urine and much water was lost through urine reducing blood volume low blood volume reduced venous return and since blood pumps resulting into lower cardiac output, blood becomes more viscous and does not easily flow to blood to the brain for t to control walking; ✓Ap

Sensory cell hair cell integrity was partly damaged in David; ✓Al due loud extreme levels of noise from the out door music festival; ✓Ae Sensory hair cells in the Cochlea High sound intensity produces energetic waves that travels through air and enters the ear through the canal to ear drum and through middle ear via ear bones to the inner ear. This vibration cause basilar membrane to vibrate which push sensory hairs against the tectorial membrane to slide past each other. The distortion of hairs due to the shearing forces causes depolarization of sensory cells

to produce generator potentials. However due to high intensity over distortion caused damage to the sensory hairs (cells) The damaged cells can spontaneous generate impulses have a feeling of "ringing in the ears";✓Ap

Endolymph fluid dynamics in semicircular canals was more turbulent than normal flow in the David; ✓Al because of Dancing and shaking the head violently and consumption of alcohol; ✓Ae alcohol changes blood and endolymph composition after diffusing into the endolymph making theendolymph less dense / altered viscosity. This abnormal turbulent flow makes the cupula to bend even if the head is not moving Causing false sensation of to the cerebellum and motor centres the cerebellum fail to control muscles well and David fails to walk in a straight line / staggers. ✓Ap

(b) Isotonic saline solution will be effective; ✓El because it stays in blood and increases blood volume which increases venous return to the heart, increasing stroke volume and cardiac output making blood to easily flow to the brain However and restores Na+ and Cl- lost in urine for proper generation of membrane potentials. However if the saline is given alone without glucose it does not provide energy for the brain cells that is low due to poor feeding and alcohol metabolism; ✓Ee Therefore continue with this method but mix glucose into the normal saline solution for brain cells to function properly; ✓Ep

Giving David oral rehydration salts will be effective; ✓El because because ORS contains Na+ Cl- glucose and water Na+ and K+ will replace the ones lost in urine after being absorbed into blood and glucose will be absorbed to increase blood sugar for brain cells to use However ORS may be easily vomited due to alcohol and due to turbulent endolymph nausea can worsen from vestibular stimulation; ✓Ee Therefore continue with this strategy after but give ORS in small frequent sips not a large volume at once to reduce stimulation to vomit;✓Ep

Quiet rehabilitation Environment will be effective; ✓El because it allows damaged hair cells to stabilise and prevent further shearing by removing the noise trauma for the remain 65% of healthy cells and damaged cells recover. However if keep in a quiet room for long without balancing exercise his brain will be slow at learning balance again since cerebellum learns balance by practice; ✓Ee Therefore continue with it but incorporate balancing exercise to make David walk again normally in a short time; ✓Ep

ITEM 3.8

(a)Plasma ADH concentration in Mark was higher than normal; ✓Al because of Heavy sweating earlier or during exercise; ✓Ap Heavy sweating resulted in loss of large amount of water resulting in blood being very concentrated; ✓Ae Concentrated blood flowing through the hypothalamus stimulated hypothalamus to stimulate release of ADH from posterior pituitary gland which travelled in blood to the kidneys where it caused massive reabsorption of water in DCT and collecting duct causing very low urine output and Dark yellow urine in Mark; ✓Ap

Core body temperature was Higher in Mark than normal; ✓Al because of sweating stopping; ✓Ae sweating provides a way of heat loss through evaporative cooling so when sweating stopped due to very low blood pressure prevented heat loss and heat accumulated in the body raising core body temperature in Mark; ✓Ap

Heart rate in mark was higher; ✓Al because of heavy exercise which produced excess CO2 lowering blood pH; ✓Ae High levels of Co2 from respiration of active muscles was detected by chemoreceptors which sent impulses via sympathetic nerves to the cardiac accelerator centre which send impulses to the Heart (SAN) to increase heart rate; ✓Ap

Plasma sodium ion concentration in mark was lower than normal; ✓Al because of heavy sweating; ✓Ae during sweating sweat is produced containing large amounts of water and sodium ions so loss of large amounts of sweat reduced the blood sodium level in mark; ✓Ap

Blood pressure was lower because in mark; ✓Al because of large amounts of water loss through sweating; ✓Ae This reduced blood volume and consequently reduced pressure of the blood. This low blood pressure reduced blood flow to the brain causing a feeling of dizziness and confusion in Mark; ✓Ap

There was lower oxygen saturation in mark; ✓Al due to heavy exercise the active muscles consumed heavy amounts of oxygen To generate the needed ATP lowering the blood oxygen saturation; ✓Ae

(b)
Administer oral fluids; ✓El This will be absorbed into blood stream to increase blood volume which will raise pressure and allow evaporative cooling through temperate sweating to reduce core temperature However excessive fluid too much fluid may dilute blood and cause brain oedema which can result to death; ✓Ep

Therefore continue with this method but insert water slowly and mix it with salts and glucose; ✓Ep

Administer oxygen through oxygen mask; ✓El This will increase diffusion of oxygen into the lungs and blood and increase oxygen saturation allowing cells to respire aerobically However high oxygen concentration may cause oxygen poisoning by reducing breathing move'ments causing CO2 retention; ✓Ee Therefore continue with this method but give oxygen mask only to treat symptoms of anoxia; ✓Ep **4.1**

(a) Percentage of resistant insects to pesticide increased from 2014 to 2024;✓ Al because of continuous pesticide application season after season;✓ Ae the pesticide application provided a selection pressure to insects. The insects that were having pesticide resistance to this pesticide were selected for / favoured while the susceptible ones were selected against, Resistant insects grew and could not die, after each pesticide application the susceptible ones would die, reducing competition leaving abundant food sources for the resistant ones. They could grow and reproduce increasing in population season after season;✓ Ap

Predatory birds decreased; ✓ Al because of bioaccumulation the pesticide being used could not be digested or excreted so it persisted in bodies of organisms that could eat contaminated insects✓ Ae This could bio accumulate/increase in concentration at higher trophic levels reaching lethal concentrations resulting into death of top carnivores like the predatory birds✓ Ap

Insect pollinators decreased from 9200 in 2014 to 3400 in 2024; ✓ Al because of using a broad spectrum pesticide to kill pests; ✓ Ae Broad spectrum pesticide is not specific to the targeted pest so it kills a wide range of insects This resulted in death of non-targeted insect pollinators resulting to the decrease in their number✓ Ap

There was increase in CO2 levels in the atmosphere from 398ppm in 2014 to 426.8 in 2024;✓ Al Because of deforestation and emission of CO2 from burning of petroleum fuels in coffee processing industry;✓ Ae Increased CO2 resulted in increased temperature since it is a green-house gas and trap the infrared radiation and prevent it from escaping back into space keeping heat near the surface✓ Ap

(b) Mass afforestation using fast growing Eucalyptus trees will not be effective at restoring biodiversity;✓ El because eucalyptus trees produce no edible fruits,

flower once a year and no nesting sites for most organisms like birds although they can grow quickly and provide shade and nesting sites to some birds and shelter for insects rather than bare cleared ground;✓ Ee Therefore abandon this mass planting of eucalyptus but instead plant mixed indigenous fast growing species✓ Ep

Complete ban of broad spectrum pesticide will be effective at reducing death of predatory birds;✓ El because it will directly stop bioaccumulation and bio magnification / reduce for contaminated food for the birds, and restore food base since the broad spectrum kills non targeted insects which may have been food for the birds however pests will increase greatly and cause economic losses to crops and the insecticide may stay in the soil and fat of insects for years after spraying;✓ Ee Therefore continue to ban broad-spectrum herbicide but introduce IPM to control pests and use pesticide which do not persist and are biodegradable;✓ Ep

Wiping out invasive species using narrow spectrum herbicide will be effective;✓ El because the herbicide rapidly kills the invasive species in a short time while causing less risk/damage to non-targeted crops however the herbicide may persist in bodies of living organism and cause bioaccumulation and death of top predators;✓ Ee Therefore continue with the method but use the herbicide that is biodegradable;✓ Ep

4.2 (a) From 2019 to 2022, Native species decreased;✓ Al This because of spread of invasive species in the wetland;✓ Ae Invasive species grow and reproduce fast and compete with native food as well as prey feed on them reducing their population;✓ Ap

Crop yields yields reduced from 2019 to 2022;✓ Al because of reduced number of pollinators and rampant soil erosion;✓ Ae Soil erosion washes away the top soil with nutrients leaving a soil less in nutrients like nitrates and phosphates which plants need for proper growth and decrease in number of pollinators reduces chances of pollen landing pollination and fruit develop / seed development leading to poor yields decline in crop yields;✓ Ap

Percentage of bacteria tolerant to anoxic conditions increased from 2019 to 2022;✓ Al because of reduced dissolved oxygen in the water from atmosphereS;✓ Ae This created selection pressure that favoured the tolerant bacteria while selecting

against non-tolerant ones The tolerant bacteria to anoxic condition therefore divided rapidly and increased in percentage;✓ Ap

The average environmental temperature increased;✓ Al due to burning of vegetation; ✓ Ae Vegetation contains carbon compounds which when oxidized during burning produces carbon dioxide into the atmosphere Carbon dioxide being a greenhouse gas, its molecules trap long infrared radiation and prevent them from escaping back into the atmosphere resulting trapping heat near the surface;✓ Ap

(b)Spraying with a broad spectrum herbicide to control Invasive water hyacinth will be effective;✓ El because the herbicide will rapidly kill many water hyacinth in a short time reducing the numbers and cover however since it is broad spectrum it may result into death of non-targeted species and may lead to bio accumulation if it persist i.e. non-biodegradable;✓ Ee Therefore, Continue and use herbicide but use a specific herbicide for targeting only water hyacinth that is biodegradable;✓ Ep

Planting fast growing exotic pine trees will be effective at reducing CO2 level;✓ El these trees w the trees will grow very fast and form a dense cover rapidly carry out photosynthesis and absorb carbon dioxide from the atmosphere. However they still take time to grow and absorb a reasonable amount of carbon dioxide;✓ Ee therefore continue planting. Therefore continue and plant the trees but accompany it with reduction in bush burning to reduce emission; ✓ Ep

planting fast growing exotic pine to promote biodiversity will not effective at restoring biodiversity;✓ El because pine does not produce edible fruits for other species, flowers once since it is monoculture and provide no nesting sites for birds, pollinators and pine litter produces chemicals that suppresses germination of native species there although it grows quickly and provides habitat to some birds and other animals;✓ Ee Therefore abandon pine planting and replace it in planting mixed indigenous plant species to promote biodiversity; ✓ Ep

4.3 (a)

From year 0 to year 20, Native bird population decreased in number;✓ Al because of introduction and increase of invasive Snake species; ✓ Ae When the Invasive snake species were introduced, Invasive species so usually reproduce very fast and

rapidly grew and feed on native birds. This increased predation resulted in reduction in numbers of native bird species;✓ Ap

Invasive snake population increased from in the 20 years; ✓ Al because after introduction there were abundant food in form of native birds and mammals; ✓ Ae The availability of food provided good condition for snake to feed grow and reproduce resulting to increase in numbers; ✓ Ap

Crop yield decreased in the 20 years; ✓ Al because of high pest damage;✓ Ae invasive snake species increased and reduced the bird population which were predators to the Insect pests. This reduced the predation on Insect pest resulting into their rapid population growth causing increased damage to Crops and reducing the crop yield; ✓ Ap

Native birds with long and strong beaks Increased; ✓ Al because of having strong beaks and long beaks; ✓ Ae These features made them to easily break hard dead tree logs to to reach into the few available insect larvae hence they easily outcompeted the ones with short and soft wings, grew and reproduced to increase in number after spraying; ✓ Ap

Percentage death of pests after pesticide application decreased from 90% to 30%

Indicating pest resistance; ✓ Al due to continuous use of the same pesticide;✓ Ae there was a small proportion of pesticide which were resistant or the pesticide may have caused a mutation resulting into resistance. Continuous application provided a selection pressure that favored the resistant but killed the non-resistant hence the resistant increased in numbers and contributed a large percentage so most could not die after spraying. ✓ Ap

4.3 (b) Importation of large population of mongoose will be effective at reducing invasive snake population; ✓ El because large population mongoose will kill many snakes in a short time, will not introduce harmful effects of chemicals and however the Mongoose may have alternative host and may slowly reproduce to maintain snakes under check and may need time to adapt to the new environment conditions✓ Ee Therefore the mongoose should be first studied and optimum conditions for survival and reproduction;✓ Ep

Killing the snakes poison in rat baits will be Effective at killing snakes; ✓ El because the snakes will feed on poison and die instantly in a short time however poison introduce chemicals in ecosystem after snakes die which may bio accumulate in

other organisms causing death of untargeted organism;✓ Ee therefore continue with this strategy but use a poison that does not bioaccumulate and is biodegradable; ✓ Ep

Replacing indigenous crops with poor yields and planting a single clone of GM crops with good pesticide resistance will be effective at increasing yield; ✓ El since they will not be attacked by pests and produce large quantities of food however they will result into great loss of biodiversity since most local varieties will be destroyed; ✓ Ee therefore continue to plant them but on a small piece of land and maintain all the local varieties too to ensure food availability and while maintaining local biodiversity; ✓ Ep

4.4 (a)

Resistant beetles percentage increased from year 1 to year 5;✓ Al because of long term use of the same pesticides; ✓ Ae This is because continuous use of pesticide provided a selection pressure that selected for the beetles which had resistance. Therefore they were favoured and grew and reproduced as the death of susceptio ones would reduce competition, they their percentage increased;✓ Ap
There was pest resurgence;✓ Al because the number of pests increased to even greater levels; ✓ Ae This is due to the pesticide being a broad spectrum and causing far negative effects on other organisms like predators of the pest (Birds) than the pest itself. The pests multiply quickly since predators are reduced and increase over to greater numbers;✓ Ap

Number of insect eating birds decreased; ✓ Al due to bio accumulation;✓ Ae The pesticide used was a persistent, hard wide and accumulated in bodies of Birds to after eating every several contaminated insects for a long time the pesticide reached lethal level resulting to death of birds;✓ Ap

Number of pollinating Insects decreased; ✓ Al due to continuous pesticide application;✓ Ae This is due to the pesticide being a broad spectrum pesticide may end up killing non targeted species like pollinating insects hence of decline in numbers. ✓ Ap

There was allopatric speciation;✓ Al Since the beetles in separate coffee fields were separated by geographical barriers; ✓ Ae being separated/isolated into sub population and since gene flow is prevented and organisms are failing to meet may

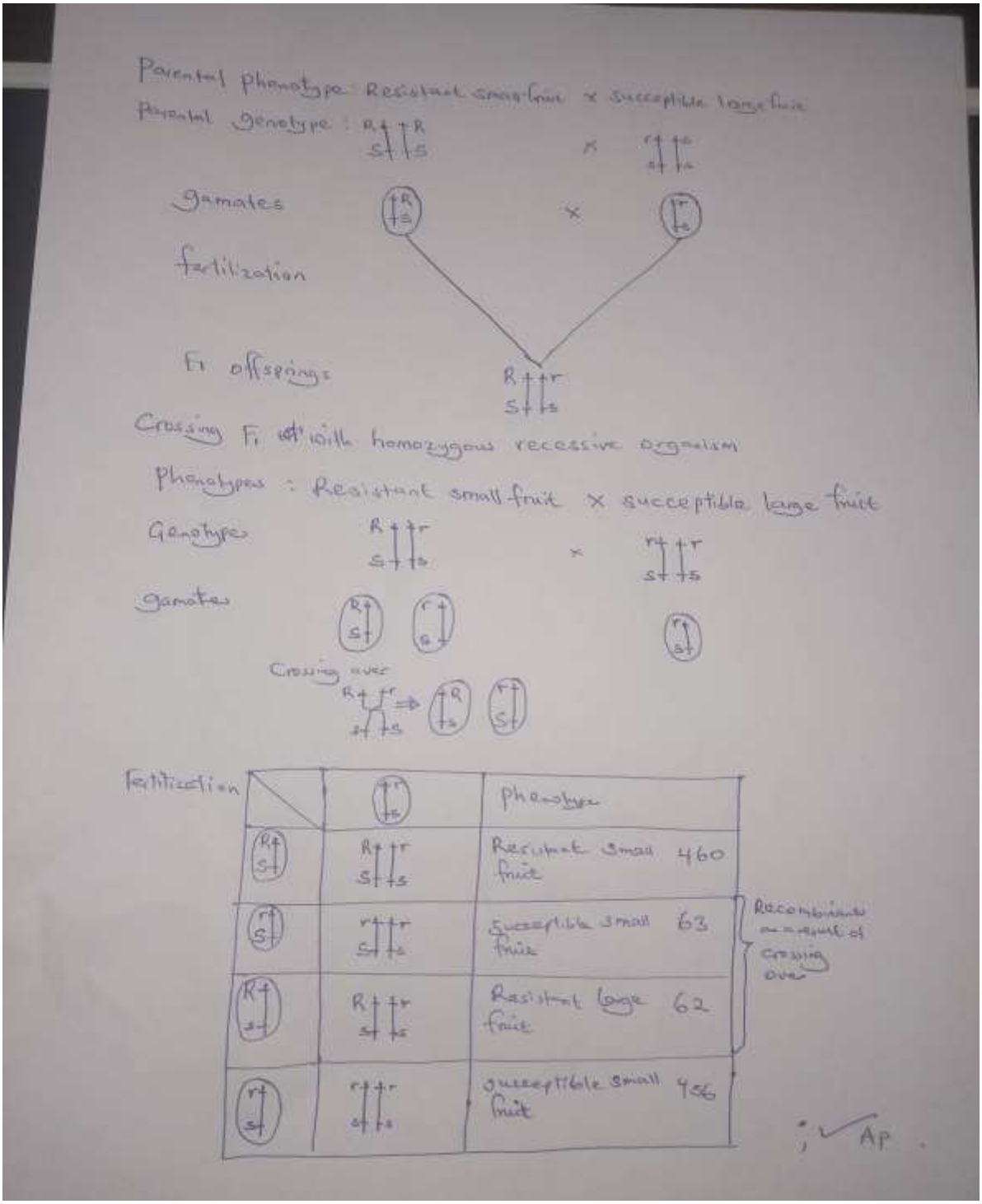
result into reproductive isolation, adaptation to the new different environments or random genetic drift in subpopulations may lead to change in allele and genotypic frequencies leading to genetic isolation hence creating new species; ✓ Ap

(b) pesticides is not the most effective way of controlling pests;✓ El because the pest can develop resistance if the pesticide is used again and again and pesticides can result into death of untargeted species and bioaccumulation and death of predators at the top of the chain however pesticides can kill many pests for short time at first and can work on pests over a wide area;✓ Ee therefore abandon the use of pesticides only and use integrated pest management;✓ Ep

Continued pesticide use can be effective at causing resistant beetle populations to evolve into separate species;✓ El because pesticide use provide directional selection to isolated populations of Beetles which favours resistant Beetles favoring increase in frequency of of such phenotypes within a population If gene flow is prevented, each sub population develops adaptation to the new environment and get mutations and eventually develop into different species of resistant beetles However if gene flow is not completely prevented, or migration keep occurring between fields speciation will not occur;✓ Ee Therefore for speciation to occur directional selection must be accompanied by prevention of gene flow between the two sub populations;✓ Ep

4.5 (a)

Resistant large fruit coffee came up as a result of crossing over; ✓ Al because there were recombinants; ✓ Ae the alleles/genes for fruit size and pest resistance are linked and in F1 there was crossing over during meiosis creating recombinant gametes.



The allele for pest resistance is dominant to that of susceptibility and allele for small fruit is dominant to that of large fruit;✓ Al because after crossing pure breeding Resistant and small fruit coffee with pure breeding susceptible with large fruit all F1 offspring were resistant with small fruit;✓ Ae

The genes for pest resistance and fruit size are linked; ✓ Ae because during test crossing the recombinants were fewer than expected and almost equal; ✓ Ae Recombinants if the genes were on separate chromosome there would be random assortment of gametes RS, Rs, rS and rs to give a ratio of 1:1:1:1 a test cross ratio for dihybrid cross however the Rs and rS gametes were very few indicating there was no independent assortment indicating that they were linked✓ Ap

The Resistant Large fruit coffee was due to crossing over✓ Al because it is showed recombination of parental phenotypes✓ Ae

4.6 a) There was allopatric speciation;✓ Al There was isolation of the rodents in 3 fields by a geographical barrier / road construction;✓ Ae These barriers restricted gene flow and this inability of organisms or their gametes to mate leads to reproductive isolation; adaptations to the new conditions or random genetic drifts in small populations lead to changes in allele and genotypic frequencies prolonged separation may lead to genetic isolation that there can no longer breed together to create fertile offspring hence forming new species;✓ Ap

Non native species rodents increased significantly while that of native rodents decreased✓ Al because they possess stronger limbs and large incisors;✓ Ae These features make them to be selected for and out compete native rodents for resources like food and nesting site resulting into high reproduction and increase in population;✓ Ap

4.6 (b)

4.7 (a) Net primary productivity was highest during light grazing; ✓ Al because the dominant plant species were perennial Themeda triandra which is highly more palatable; ✓ Ae hence consumed by primary consumers making most of the energy absorbed from the sun to be transferred to secondary consumers✓ Ap

Soil erosion was highest during heavy grazing; ✓ Al because the dominant vegetation was Sporobolus pyramidalis which does not cover the soil well like grasses; ✓ Ae hence soil easily washed away by agents of soil erosion i.e. running water; ✓ Ap

The cattle genotype of T^ET^E and T^ET^S had very low population frequency; ✓ Al because of producing low mean daily milk production; ✓ Ae These genotypes

although very tolerant to drought were selected against by farmers due to their low milk products hence their numbers remain low in the population; ✓ Ap
The cattle of genotype T^ST^S had a low frequency;✓ Al because of being very susceptible to drought✓ Ae Although these cattle had very high daily milk production they were highly sensitive to drought and could easily die out during periods of drought / selected against by farmers; ✓ Ap

The cattle of genotypes T^MT^M and T^MT^S had high population frequencies;✓ Al because they were tolerant to drought and gave reasonable milk products;✓ Ae These qualities gave them heterozygous advantage than homozygotes of either type hence were selected for reproduced and multiplied in numbers hence high population frequencies;✓ Ap

(b) Clearing the weed using a broad spectrum herbicide will not be effective;✓ El This is because the broad spectrum herbicide will kill the weed and grasses in a short time removing competition for native species to grow since it is nonspecific however it will result into elimination of the weed;✓ Ee Therefore abandon the method and use a highly specific herbicide to kill only the weed without affecting untargeted species; ✓ Ep

putting paddocks will highly be effective; ✓ El because paddocks keeps cattle in one place allowing adequate recovery time for native grass in other paddocks however if not actively balanced rotation animals may destroy completely grass in one paddock and is affected by weather changes;✓ Ee therefore continue with paddocking but delivery balance rotation timing with actual grass growth and weather changes;✓ Ep

Culling the animals of cattle showing T^ST^S genotype will be effective;✓ Al because it will remove animals susceptible to drought leaving cattle some that can withstand the available condition however those alleles give animals high milk production; ✓ Ee therefore inbreed these cattle with those showing high drought resistance to ensure drought resistant with high milk production / heterozygote advantage;✓ Ep

4.8

(a) The genes for pest resistance and the gene for wing morphology are linked; ✓ Al because of Ratios obtained after test cross not being expected 1:1:1:1;✓ Ae The two

genes did not assort independently according to Mendel's second law and gave only parental characteristics with very few recombinants that expected at low temperatures✓ Ap

Recombinants of Resistant wrinkled wings and susceptible long wings came as a result of crossing over;✓ Al because there were recombination of parental characteristics;✓ Ae during prophase (I) of meiosis during gamete formation in heterozygous F_1 the following gametes are formed.
At high temperatures there is reduced chances of crossing over;✓ Al because there are no recombinants at 32°C✓ Ae This will maintain the harmful linkage group of long wing and Resistant to pesticide which makes the pest hard to control since they have long wings to escape predators and cannot be killed by pesticides

From year 1 to year 10, frequency of the Resistant allele R increased;✓ Al because it makes the moth resistant to pesticides;✓ Ae When the pesticides are applied, the susceptible mosses die reducing competition for resistant ones which grow and reproduce and increase in the next generation leading to increase in the Resistant allele R;✓ Ap

Predator population per square km decreased from year 1 to year 10; ✓ Al because of continuous use of synthetic pesticides; ✓ Ae This is because the pesticides killed most of their prey resulting into shortage of food hence predators died of starvation✓ Ap

(b) Application of highly selective pesticides will be effective at restoring the ecosystem; ✓ El because it will result into death of only targeted pests of cabbage leaving other untargeted organisms however killed pests may be food to other organisms in the ecosystem; ✓ Ee therefore alternative food sources should be introduced after pesticide application✓ Ee

Introduction and conservation of predatory beetles will be effective at restoring the ecosystem; ✓ El because it will result into death of reduction of pests without introducing chemicals and no risk of bioaccumulation while maintaining diversity however predatory beetles may turn invasive and kill other non-targeted prey resulting into disruption of the ecosystem; ✓ Ee Therefore only native beetles should be introduced to avoid invasive species; ✓ Ep

Practicing crop rotation will be effective at controlling this pests; ✓ Elbecause when crops are rotated, lifecycles of pests are broken reducing pest population however it will only be effective if the pest cannot attack successive crops; ✓ Ee therefore plant crops used should not be attacked by the pest✓ Ep

4.9

(a) From 1960 to 2026 the Nile perch population increased;✓ Al because they are native invasive species; ✓ Ae These rapidly reproduced and had abundant food in form of native species and outcompeted the native fish for resources like prey resulting into increase in number✓ Ap

Number of active this cichlid fish was highest before introduction of invasive nile perch and decreased after introduction of nile perch; ✓ Al because of introduction of nile perch✓ Ae Nile perch are invasive species and easily multiplied and feed on native fish and outcompeted the native fish for resources hence decrease in number due to extreme starvation; ✓ Ap

From 1980 to 2026 water hyacinth cover percentage increased; ✓ Al because it is an invasive species; ✓ Ae therefore undergoes rapid reproduction out competing native weeds to cover most part of water surface✓ Ap

Frequency of G-gene that controls large gill surface area increased; ✓ Al because water hyacinth cover✓ Al because increased✓ Ae high water hyacinth cover prevent light penetration limiting photosynthesis by phytoplanktons reducing dissolved oxygen and also they die and decompose by aerobic bacteria which deplete oxygen therefore low dissolved oxygen produces a selection pressure that favors fish with gills with large surface area to increase oxygen uptake hence they increase in number✓ Ap

(b) Breeding cichlid species in controlled highly oxygenated nile perch free environment will be effective✓ El and because the number of these native fish will increase and return to normal population however when re-introduced in the lake, nile perch will again reduce their population to critical levels through predation✓ Ee therefore continue with this method but accompany it with controlling nile perch species by selectively fishing them and selling them out; ✓ Ep

Spraying water hyacinth with herbicides will be effective; ✓ El because it will kill water hyathinth reducing their population and removing the cover allowing light penetration and restoring dissolved oxygen however chemicals may bioaccumulate resulting to death of top predators; ✓ Ee therefore only biodegradable herbicides should be used;✓ Ep

Introduction of specific biological agents in the lake to reduce water hyacinth and will be effective;✓ El because biological agents will feed and reduce the water hyacinth and allow light penetration restoring dissolved oxygen however the biological pest control used may have alternative prey / reproduce slowly to remove the hyacinth quickly as needed; ✓ Ee hence a biological agent that reproduces very fast and only targets water hyacinth should be used;✓ Ep

4.10 (a)

From 1996 to 2008 algal biomass increased; ✓ Al because of cultivation on surrounding hillsides;✓ Ae In the fields where cultivation is being done, Nutrients like nitrates and phosphates are washed by surface soil erosion running water into the lake. This results in nutrient enrichment and stimulates growth of algae hence increase in algal biomass;✓ Ap

From 1996 to 2008 dissolved oxygen decreased;✓ Al because of rapid growth of algae; ✓ Ae when algae increase, they prevent light penetration into the lake resulting into death of algae that are obstructed from light. They die and are decomposed by aerobic bacteria which increase and use up the dissolved oxygen hence decrease in dissolved oxygen;✓ Ap

Frog population decreased from 1996 to 2008; ✓ Al because of low / decreasing dissolved oxygen✓ Ae Frogs tadpoles are aquatic and depend on dissolved oxygen to survive this low dissolved oxygen increased mortality of tadpoles. Also decomposing algae produce ammonia and hydrogen sulphide which are toxic to tadpoles hence decrease in population✓ Ap

(b)

Introduce zooplankton and herbivorous fish into the lake;✓ El This will help to reduce algae by feeding on them hence reduce their decomposition and reduction in dissolved oxygen however these fish may become invasive and multiply

uncontrollably and affect the ecosystem; ✓ Ee therefore introduce fish that are well studied not to outcompete native species of fish; ✓ Ep

Control soil erosion from the hills by terracing and establishing buffer zones at 50m uncultivated buffer strip of grasses;✓ El This will reduce nutrient / sediment run off into the lake and reduce eutrophication however not all nutrients may be absorbed and they will end up into the lake;✓ Ee Therefore this should be accompanied with abandoning use of fertilisers and use organic manure instead;✓ Ep

manual harvesting of floating blooms;✓ El this will increase light penetration allowing photosynthetic algae to photosynthesize and produce oxygen reducing decomposition however if nutrients are there the algae will always bloom again✓ Ee therefore this method should always be repeated after specific time intervals✓ Ep